

67. TRAINING AT FATMAX WITHOUT ANY LAB TEST

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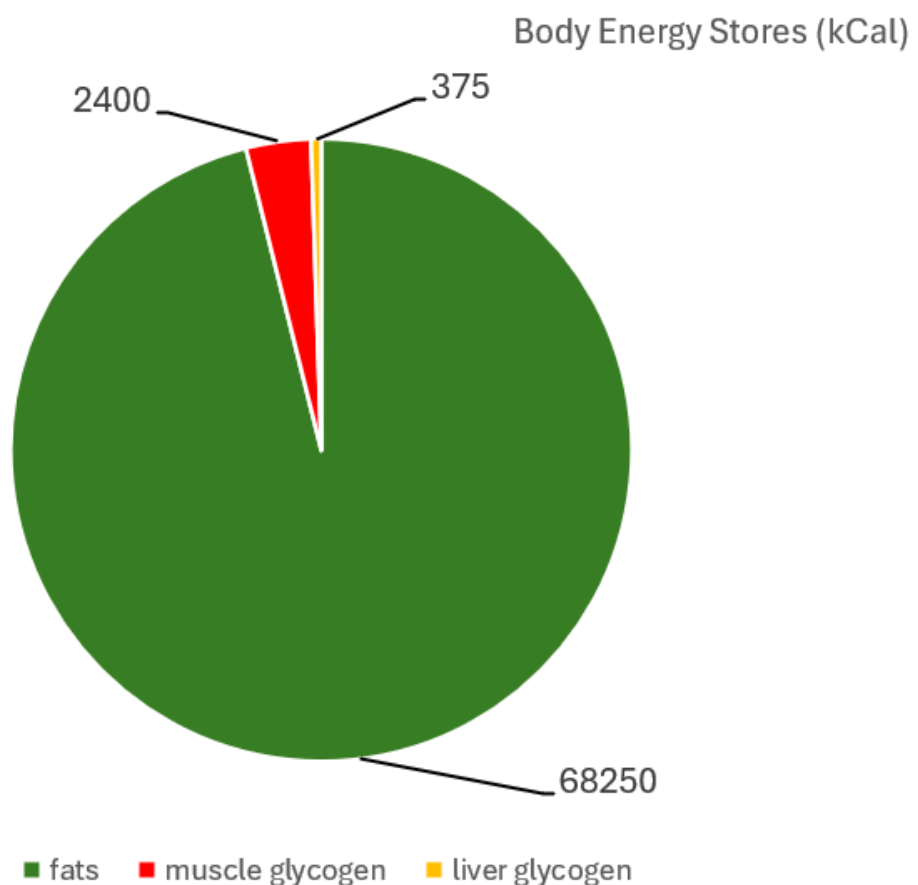


'Fatmax is the intensity at which the maximum rate of fat oxidation is elicited'.

This is usually expressed in power output or heart rate range (for example 220-240 W or 130-140 bpm). Many cyclists want to know their fat max to train at this intensity. Why? The training specificity principle states that closer the training routine to the desired outcome, better the adaptations obtained. Following this principle, if we train at fatmax intensity we are stimulating maximally the fat metabolism and this could improve the ability to use fats as fuel during exercise and so durability.

Why training at fatmax and becoming better to use fats as fuel could improve durability?

[We saw](#) that durability is linked (in part) to preserve muscle glycogen levels and carbohydrates availability. **Even in very lean individuals, fats are a virtually unlimited energy store while carbs are not.**



Energy stored (kcal) in the body. Calculations are based on Maunder et al. 2018 (1) and consider a 70kg man with a 10% body fat

So, being capable to use as much as possible fats (the unlimited resource) during exercise could make spare carbohydrates/glycogen (the limited resource), and increase durability.

It is important to acknowledge that, while this approach seems intuitively reasonable and widely used on the field, no study has yet demonstrated this.

In a previous 'in depth' post (read more [here](#)), we saw that **fatmax should occur approximately at ~60-70% FTP/second threshold or 65-75% Heart Rate max**. How did we make this estimation? From a big dataset (2) used by Ed Maunder and Colleagues (Auckland University of Technology) which highlighted that in 201 lean trained males fatmax was elicited at 56 ± 8 % of VO2max. Considering that VO2max is ~120% of power at second threshold/FTP, with some simple math we derived that:

Fatmax = ~60-70% FTP/second threshold or 65-75% Heart Rate max

However this was just a calculation without a direct measure. Does it really hold true?

Can we be sure that we are training at Fatmax without any lab test?

Data from a hot off the press study published by Meixner and Colleagues (University of Würzburg) on *Translational Sports Medicine* in 2025 can help to answer this question. (1)

WHAT DID THEY DO?

- They recruited 50 trained amateur cyclists (VO2max: 54)
- The cyclists performed an incremental step test (100 W + 25 W every 3min) to detect and compare the intensities (power) at which the following physiological markers occur:
 - **Fatmax**
 - **72% of Maximum Heart Rate**
 - **82% of Maximum Heart Rate**
 - **First Ventilatory Threshold (VT1)**
 - Blood Lactate Concentration = 1.5 mmol/L
 - Blood Lactate Concentration = 2mmol/L
 - Blood Lactate Concentration = 2.5 mmol/L
 - Blood Lactate Concentration = baseline + 0.5 mmol/L

Before the incremental test started, participants ingested 35 grams of carbs, so the measurements were taken in a similar nutritional scenario of a training session on the road.

WHAT DID THEY FIND?

Here the differences between the power output at first ventilatory threshold (VT1) against the other markers:

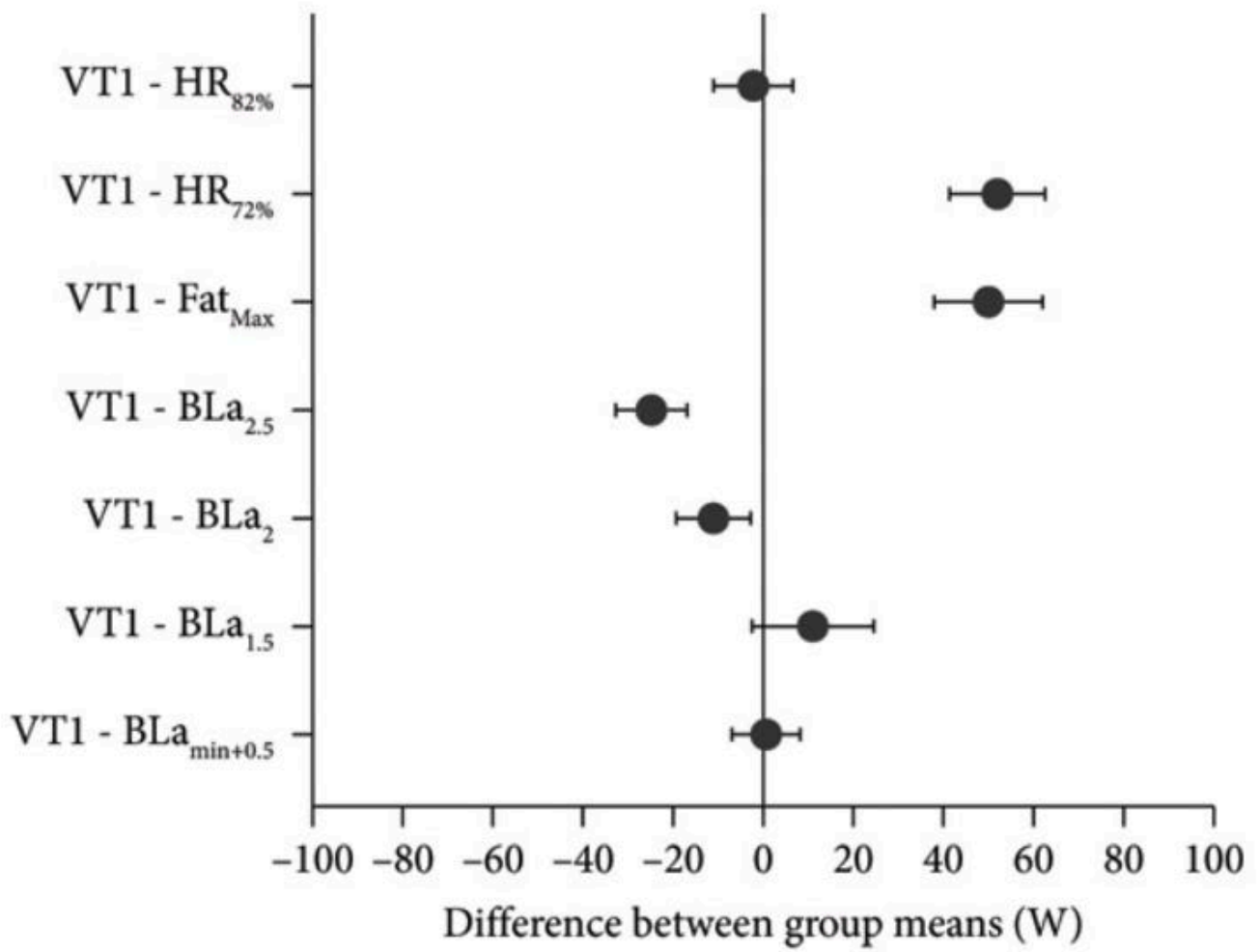


Figure from Meixner et al. 2025 (1).

Here the difference between power output at Fatmax against the other markers:

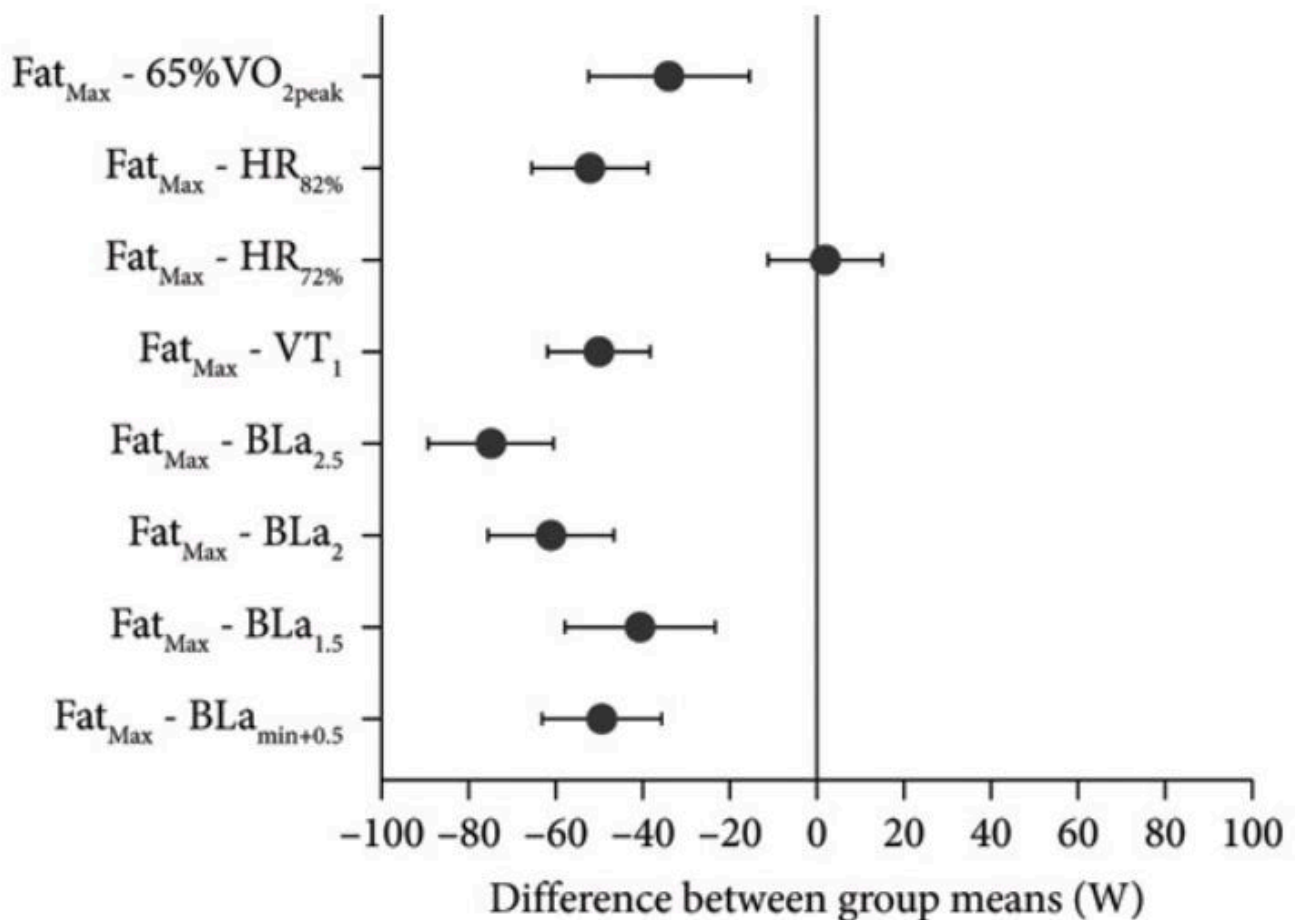


Figure from Meixner et al. 2025 (1).

PRATICAL TAKEAWAYS:

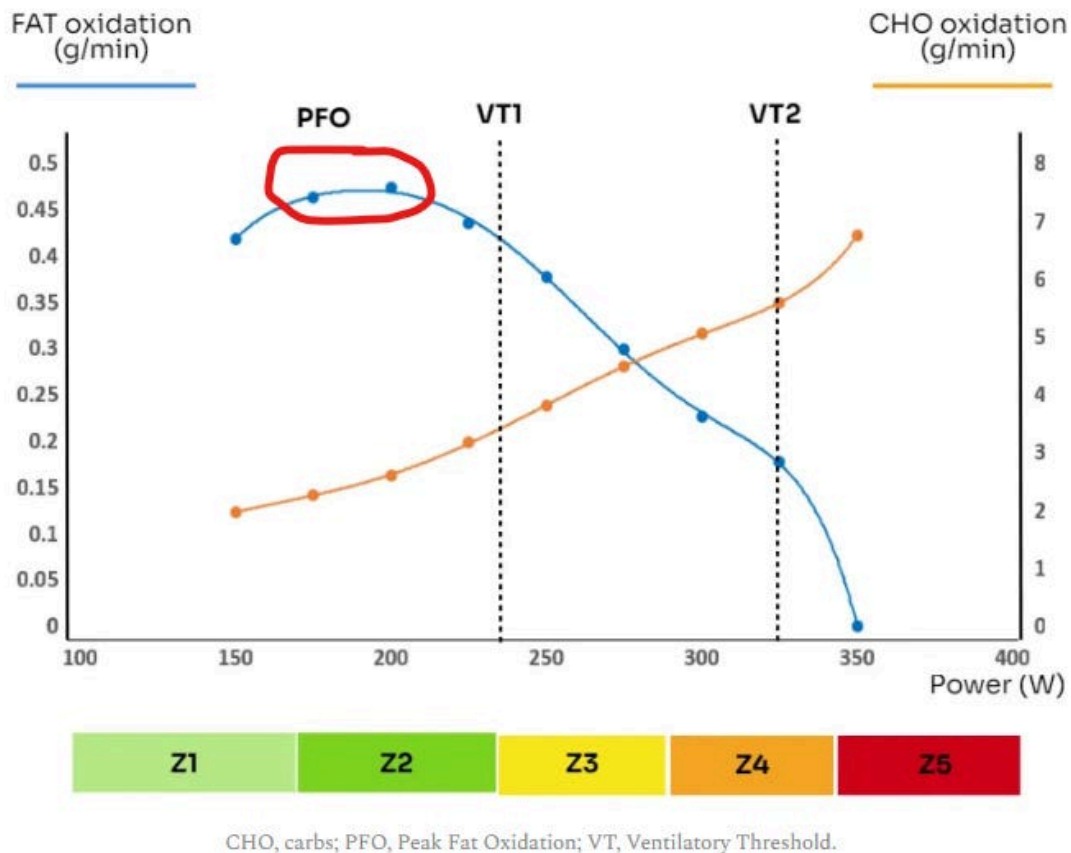
Looking at the above data, you can see that, on average, in amateur trained cyclists:

- VT1 is pretty close to 2mmol/L of blood lactate and 82% of heart rate max, with **baseline +0.5 mmol/L of blood lactate being the closest measure to VT1**.
- However, **Fatmax occurs at a lower value intensity than VT1 and it is closer to 72% of heart rate max.**

GABRIELE'S COMMENT

The practical takeaways pointed above are already self-explanatory. While some interindividual differences were found, on average fatmax occurs at an intensity lower than the first threshold (or 82% of hr max), while it is close to 72% of heart rate max. So, the 65-75% of heart max or the 60-70% of FTP that we deducted and suggested using the data from Maunder et al. (1) seems pretty accurate. A range and not just 72% of heart rate max can be used to plan training at fatmax, because before VT1 (when lactate starts to accumulate) fat oxidation is still pretty stable on high levels even just below or just above fat max, while after the first threshold. it

starts to dramatically decrease. This is well described by the below figure which represents a lab incremental metabolic test I performed which directly measured the relationship between fat oxidation and exercise intensity.



Anecdotal, fatmax of elite athletes can occur closer to VT1 (in some cases even a little bit higher than VT1), so caution should be taken when copy and paste this result on professional athletes.

Thanks for reading! Share with your Team!

Gabriele Gallo, PhD in Exercise and Sport Sciences

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1. Maunder E, Kilding AE, Plews DJ. Substrate Metabolism During Ironman Triathlon: Different Horses on the Same Courses. *Sports Med.* 2018 Oct;48(10):2219-2226.
2. Maunder E, Plews DJ, Kilding AE. Contextualising Maximal Fat Oxidation During Exercise: Determinants and Normative Values. *Front Physiol.* 2018 May 23;9:599.
3. Meixner B, Filipas L, Holmlber H, Sperlich B. Zone 2 Intensity: A Critical Comparison of Individual Variability in Different Submaximal Exercise Intensity Boundaries. *Translational Sports Medicine.* 2025 Feb.



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